Microgrids at Scale:
Resilient Power Infrastructure for AI Data Centers

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Booth #B5



Delta Microgrid For AI Data Centers

- Maintain power stability with real-time balancing
- Maximize energy resource efficiency
- Ensure ongoing operation of mission-critical loads
- Minimize outage risk and downtime

Typical Microgrid Architecture

Requires an integrated, coordinated system designed for AI load smoothing and customized to support 800VDC or 480VAC architectures.



Generation Sources

Firm and clean energy generation solutions combining solar PV, wind, gas turbines, and solid oxide fuel cells.



Energy Storage

Large-scale, grid-forming battery energy storage system (BESS).



Grid and Load Interface

Utility interconnection, substations, feeders, and critical loads.

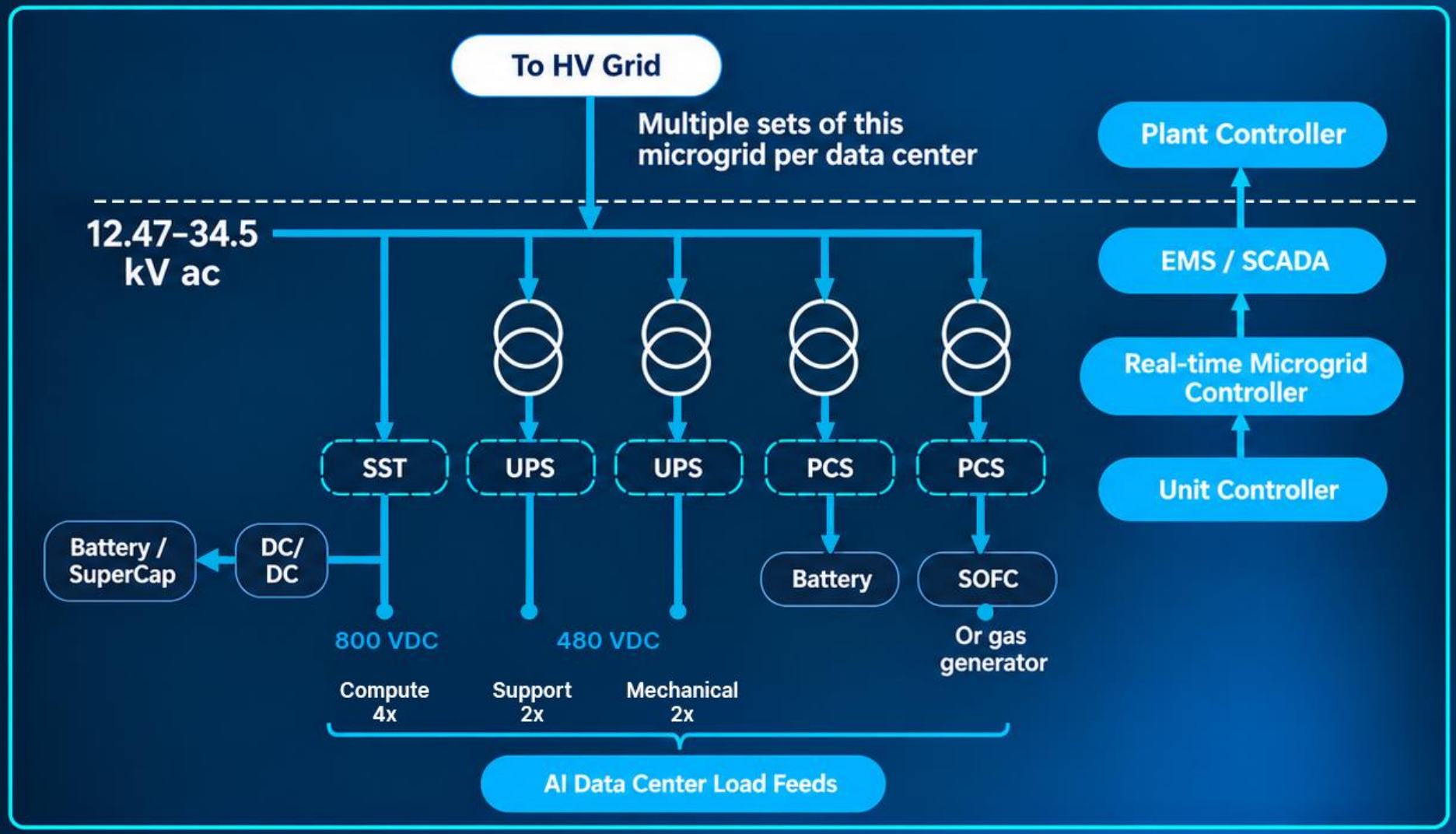


Control and Management Layers

Power plant controller, real-time microgrid controller, and EMS/SCADA platform.

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Microgrid Architecture & Components



Management Intelligence

- Forecast
- Power Dispatch and Optimization
- Grid Compliance

Edge Control

- Microgrid Control
- Power Management

Connected Products

- One unit controller per SST, UPS, PCS, etc.

The Intelligence Behind Data Center Microgrids

Delta's multilayer control architecture ensures stable, grid-compliant operation by cleary defining control authority across device-level, plant-level, and real-time systems.

	Microgrid Controller	Microgrid EMS
Stability	Voltage Regulation Frequency Regulation	Dynamic GFM, GFL DIstribution
Impedance Management	Virtual Impedance Injection and Mode Switching	SCR Prediction and Stability Maps
Inertia Management	df/dt Measurement and Allocation	Total Inertial Reserve Dispatch
AI Load Sync	Fast Load Sharing	Load Curve and Task Feed-Forward



The Delta Microgrid Advantage

Solid State Transformer: Fast Time-to-Power and Smart Architecture

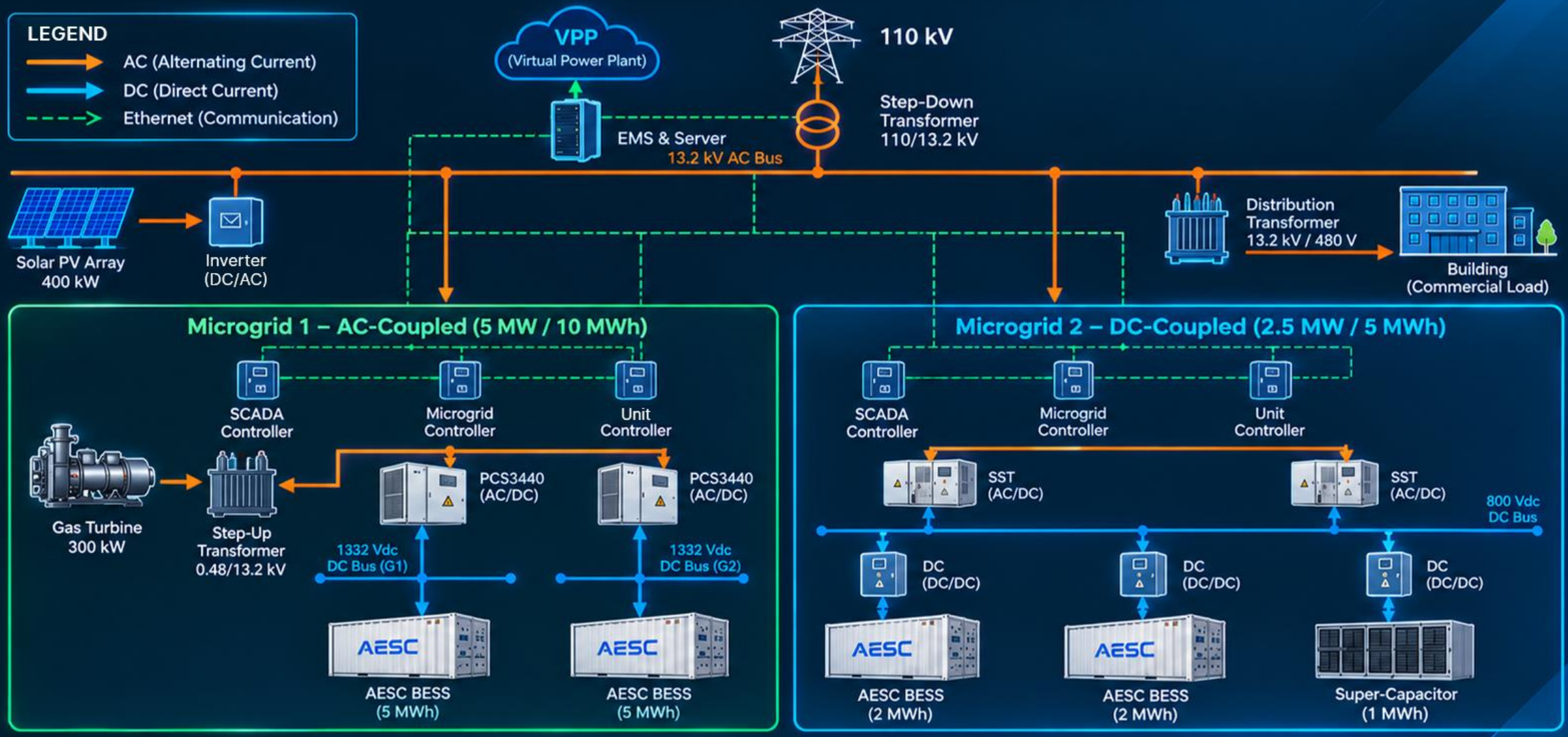
- Direct 800VDC integration
- Prefabricated and factory-verified
- Scalable, modular, small-footprint design
- 98.5% MW-to-LV conversion efficiency

Power Conversion System: Real-Time Stability and Bidirectional Control

- Real-time voltage/freqency and charge/discharge command execution
- State-of-charge and status reports through the controller

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Detroit Microgrid Demo Testing Site



For over 50 years, Delta Electronics has led the world in power electronics, automation & infrastructure.



Taipei, Taiwan
Global Headquarter

$17.9B
2025
Global Revenue

81K+
Employees

12K+
R&D Engineers

18K+
Patents

8% ~ 10%
R&D Spend (% of Revenue)

Local Manufacturing Expansion

U.S.-Based Manufacturing

Energy Infrastructure

- NEVI- and BABA-Compliant EV Charger
- EV Charger Power Module Assembly
- Single-Phase Residential PV Inverter
- Power Conversion System (PCS)
- Solid State Transformer (SST)

Data Center

- Air-Assisted Liquid Cooling (AALC) Systems
- Liquid-to-Liquid Cooling Systems
- PDU Cabinets

Telecom Power Systems

- Assembly-Wiring-Testing (AWT) for Indoor Power System
- Integrate Outdoor Power Cabinet
- Assembly of Large DC Power System
- Kitting of Small DC Power Systems

Delta's Local Repair Service Center

ISO9001 & UL Certified

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Smarter. Greener. Together.

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OCP GLOBAL SUMMIT2026

Superior Efficiency, Shaping Sustainable AI

Speed Time to Power and Reliably Operate at Lower Cost with Delta Technologies from Modular, Integrated Grid-to-Chip Architecture to Proven Microgrid, 800 VDC, and Liquid-Cooling Solutions.

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